

Lecture 3: Exponential Functions

So far every function has been built from powers of x with a *fixed* exponent — x^2 , x^3 , $1/x$. An **exponential function** reverses the roles: the base is fixed and the variable sits in the exponent,

$$f(x) = b^x, \quad b > 0, b \neq 1.$$

Exponential functions model growth or decay whose *rate* is proportional to the current amount — populations, radioactive decay, and compound interest all follow this pattern — and they are the subject of this lecture.

1. Basic Properties: The Laws of Exponents

The addition rule for positive integers. Begin with b^n as repeated multiplication when n is a positive integer. For positive integers m, n , multiplying powers of the same base adds the exponents:

$$b^m b^n = b^{m+n}.$$

Extending the rule to zero. We want the addition rule to keep working when one exponent is 0. Since $b^m b^0$ must equal $b^{m+0} = b^m$, define

$$b^0 = 1.$$

Extending the rule to negative integers. We also want $b^m b^{-m} = b^{m+(-m)} = b^0 = 1$. Thus define

$$b^{-m} = \frac{1}{b^m}.$$

The addition rule now holds for all integer exponents. Combining it with $b^{-n} = 1/b^n$ gives the quotient rule,

$$\frac{b^m}{b^n} = b^m \cdot b^{-n} = b^{m+(-n)} = b^{m-n},$$

and repeated multiplication gives the power-of-a-power and power-of-a-product rules for integer n ,

$$(b^m)^n = b^{mn}, \quad (bc)^n = b^n c^n.$$

The restriction $b > 0$ will matter when we extend exponents beyond the integers: it ensures roots such as $b^{1/2}$ are real and positive.

Example. Simplify $\frac{2^{3x} \cdot 2^{-x+1}}{4^x}$.

Write $4^x = (2^2)^x = 2^{2x}$, then combine exponents in the numerator and subtract:

$$\frac{2^{3x} \cdot 2^{-x+1}}{2^{2x}} = \frac{2^{3x+(-x+1)}}{2^{2x}} = \frac{2^{2x+1}}{2^{2x}} = 2^{2x+1-2x} = 2^1 = 2.$$

Exercise 1 (routine). Simplify each using the addition and quotient rules:

- $\frac{2^5 \cdot 2^{-7}}{3^{x+2}}$
- $\frac{3^{x-1}}{5^{2x} \cdot 5^{-3}}$
- $\frac{5^{2x} \cdot 5^{-3}}{5^{x-4}}$

Exercise 2 (algebraic). Simplify each completely:

- $(2^x)^3 \cdot 2^{-2x}$
- $(3^x)^2 \cdot 3^{-x+1}$

- $\frac{(2^{2x})^{-1} \cdot 2^{x+5}}{4^{x-1}}$

Exercise 3 (equations). Solve for x in each. (Use that $b^x = b^y$ implies $x = y$ when $b > 0$, $b \neq 1$.)

- $5^{2x-1} = 5^{x+4}$
- $2^{3x} = 2^{x+8}$
- $4^{x+1} = 8^{2x-1}$ (write both sides as powers of 2 first)

2. The Graph of $y = 2^x$ and $y = (1/2)^x$

Section 1 only defines b^x when x is an integer — the addition rule manipulates exponents, but never says what $2^{1/2}$ is. Before graphing $y = 2^x$ as a curve over *every* real x , pause on that gap.

Question. Using the power-of-a-power rule $(b^x)^y = b^{xy}$, what value must $2^{1/2}$ take so that $(2^{1/2})^2 = 2^1 = 2$? What value must $2^{1/3}$ take so that $(2^{1/3})^3 = 2$? Based on these, what should $2^{p/q}$ mean, for positive integers p, q ?

This pins down b^x for every *rational* exponent $x = p/q$: it must be $\sqrt[q]{b^p}$, the q -th root of b^p . But pinning down b^x for an *irrational* exponent — such as $2^{\sqrt{2}}$ — is genuinely harder: there is no fraction p/q equal to $\sqrt{2}$, so no finite root captures it directly. The rigorous definition approximates $\sqrt{2}$ by closer and closer rationals p/q (e.g. 1, 1.4, 1.41, 1.414, ...) and defines $2^{\sqrt{2}}$ as the number that $2^{p/q}$ approaches — a **limit**, a tool beyond this course. For the rest of this lecture we take for granted that b^x is defined for every real x , and that it varies continuously — its graph has no holes or jumps. The exponent laws above continue to hold for rational exponents and, with this continuous extension, for all real exponents.

Exercise 4 (rational exponents). Evaluate each exactly, then rewrite any negative exponent as a reciprocal:

- $16^{3/4}$
- $27^{-2/3}$
- $81^{1/2}$

Basic information for $y = 2^x$.

- Domain: $(-\infty, \infty)$ — every real exponent is allowed.
- Range: $(0, \infty)$ — a positive base to any real power is positive, and every positive value is attained.
- y -intercept: $2^0 = 1$, so the graph passes through $(0, 1)$. There is no x -intercept, since $2^x = 0$ has no solution.
- Horizontal asymptote: $y = 0$. As $x \rightarrow -\infty$, $2^x \rightarrow 0^+$ but never reaches 0.
- Monotonicity: since $2 > 1$, 2^x is increasing on all of $(-\infty, \infty)$ — larger exponents of a base bigger than 1 give larger values.

The function $y = (1/2)^x$. Using the laws of exponents,

$$\left(\frac{1}{2}\right)^x = \frac{1}{2^x} = 2^{-x}.$$

Replacing x by $-x$ reflects a graph across the y -axis, so $y = (1/2)^x$ is the mirror image of $y = 2^x$ across the y -axis. It shares the same domain $(-\infty, \infty)$, range $(0, \infty)$, y -intercept $(0, 1)$, and horizontal asymptote $y = 0$, but since $0 < 1/2 < 1$, it is **decreasing**: as x increases, $(1/2)^x$ shrinks toward 0.

Example. Compare 2^x and $(1/2)^x$ at $x = -2, 0, 3$:

x	2^x	$(1/2)^x$
-2	1/4	4
0	1	1
3	8	1/8

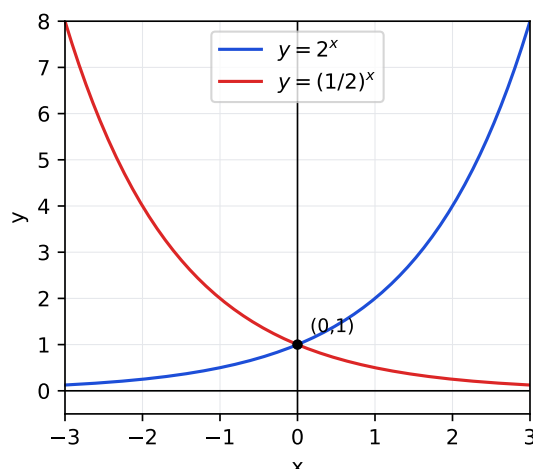


Figure 1: y equals 2 to the x and y equals $(1/2)$ to the x , reflections of each other across the y -axis, both with horizontal asymptote $y=0$

Every row confirms the mirror symmetry: swapping the sign of x swaps the two columns.

Exercise 5 (features). State the domain, range, y -intercept, and horizontal asymptote of $y = 5^x$, and say whether it is increasing or decreasing.

Exercise 6 (comparison). Without a calculator, decide which is larger: $3^{0.4}$ or $3^{0.5}$? Which is larger: $(1/3)^{0.4}$ or $(1/3)^{0.5}$? Explain using monotonicity.

Exercise 7 (sketch). Sketch $y = (1/3)^x$ on the same axes as $y = 3^x$. Label their common point, the horizontal asymptote, and the points with $x = 1$ and $x = -1$ on each graph.

3. The Number e and Continuous Compounding

A concrete case: 100% interest. Suppose \$1 is invested at a 100% annual interest rate, and it is compounded **twice** a year: every half-year the balance is multiplied by $1 + \frac{1}{2}$. After one year,

$$A = \left(1 + \frac{1}{2}\right)^2 = 1.5^2 = 2.25.$$

Compounded **four** times a year, every quarter multiplies the balance by $1 + \frac{1}{4}$, so after one year,

$$A = \left(1 + \frac{1}{4}\right)^4 = 1.25^4 \approx 2.4414.$$

Compounding more often increased the balance ($2.25 \rightarrow 2.4414\dots$), since interest starts earning interest sooner. Compounding n times a year gives $A = \left(1 + \frac{1}{n}\right)^n$; does this keep growing without bound as $n \rightarrow \infty$, or does it level off?

n (times per year)	$\left(1 + \frac{1}{n}\right)^n$
2	2.25
4	2.4414
12 (monthly)	2.6130
365 (daily)	2.7146
10000	2.7181

The balance keeps growing but the growth slows dramatically, leveling off near 2.71828.... This limiting value is called e :

$$e := \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \approx 2.71828\dots,$$

an irrational constant, already sandwiched between 2 and 3 in the table above.

The 100% example above is the case $r = 1$, $t = 1$: $A = Pe$, matching the table's approach toward 2.71828... as compounding gets more frequent. The same formula $A(t) = Pe^{rt}$ also models continuous decay when $r < 0$; the limiting argument above focuses on the positive-interest case.

Basic information for $y = e^x$. Since $2 < e < 3$, the graph of $y = e^x$ sits between $y = 2^x$ and $y = 3^x$ for $x > 0$ (and between them the other way for $x < 0$, since all three still meet at $(0, 1)$). Like every exponential with base greater than 1, $y = e^x$ has domain $(-\infty, \infty)$, range $(0, \infty)$, y -intercept $(0, 1)$, horizontal asymptote $y = 0$, and is increasing everywhere.

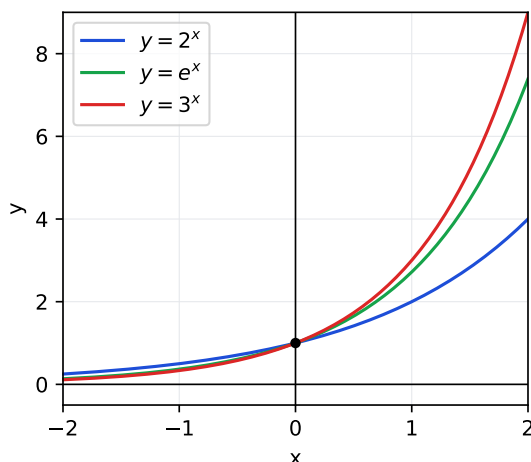


Figure 2: y equals e to the x sandwiched between y equals 2 to the x and y equals 3 to the x , since 2 is less than e is less than 3

Example. \$1000 is invested at 5% annual interest, compounded continuously. After 10 years the balance is

$$A(10) = 1000 e^{0.05 \cdot 10} = 1000 e^{0.5} \approx 1000(1.6487) \approx \$1648.72.$$

Example (credit card debt — a cautionary tale). Credit cards commonly charge around 20% annual interest, compounded *daily* — compounding 365 times a year is already so close to continuous compounding that $A(t) = Pe^{0.20t}$ is a good approximation. If a \$2000 balance is never paid down, after 1 year it grows to

$$A(1) = 2000 e^{0.20} \approx 2000(1.2214) \approx \$2442.80,$$

and after 5 years,

$$A(5) = 2000 e^{0.20 \cdot 5} = 2000 e^1 \approx 2000(2.71828) \approx \$5436.56,$$

nearly *tripling* the original debt. The same exponential growth that makes continuously compounded savings attractive makes continuously compounded debt dangerous: interest accrues on the unpaid interest itself, so an untouched balance grows exponentially, not linearly, the longer it sits. **Do not carry credit card debt** — pay the statement balance in full each month whenever you can, so no interest accrues at all.

Exercise 8 (routine). \$500 is invested at 4% annual interest, compounded continuously. Write the formula for $A(t)$ and find the balance after 20 years (leave the answer in terms of e , then approximate using $e^{0.8} \approx 2.2255$).

Exercise 9 (comparison). \$1000 earns 6% annual interest for 5 years. Write the balance if it is compounded quarterly, then write the continuous-compounding balance. Use a calculator to approximate both amounts to the nearest cent and state which is larger.

Exercise 10 (reasoning). Explain, using the formula $A(t) = P(1 + r/n)^{nt}$ and the definition of e , why continuous compounding gives a *larger* balance than compounding once a year, for the same P , r , and $t > 0$.

4. Transformations of Exponential Functions

The transformation table from Lecture 2 applies to $y = b^x$ exactly as it does to any parent graph: if (a, c) lies on $y = b^x$, then

New function	Point on new graph	Effect
$b^x + k$	$(a, c + k)$	shift up k
b^{x-h}	$(a + h, c)$	shift right h
$-b^x$	$(a, -c)$	reflect across x -axis
b^{-x}	$(-a, c)$	reflect across y -axis
$A \cdot b^x$	(a, Ac)	vertical scale by A

The most important consequence is what happens to the asymptote: since $y = b^x$ has horizontal asymptote $y = 0$, the shifted function $y = A \cdot b^{x-h} + k$ has horizontal asymptote $y = k$ — vertical shifts move the asymptote, but horizontal shifts, reflections, and vertical scaling do not.

Example. For $y = 2^{x-3} + 1$: starting from $y = 2^x$ (asymptote $y = 0$, through $(0, 1)$), shift right 3 and up 1. The asymptote moves to $y = 1$, and the point $(0, 1)$ moves to $(3, 2)$. The y -intercept is

$$2^{0-3} + 1 = 2^{-3} + 1 = \frac{1}{8} + 1 = \frac{9}{8}.$$

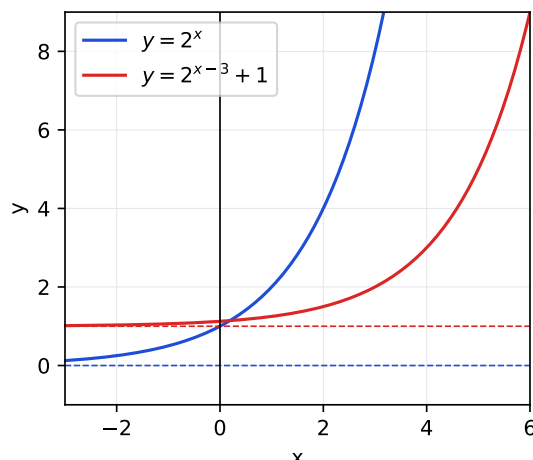


Figure 3: y equals 2 to the x shifted right 3 and up 1 to give y equals 2 to the $(x-3)$ plus 1, with asymptote moving from $y=0$ to $y=1$

Exercise 11 (features). For $y = 3 \cdot 2^x - 4$: state the horizontal asymptote, the y -intercept, and whether the graph is increasing or decreasing.

Exercise 12 (transformations). For $y = -2^{x+1} + 3$: describe the transformations from $y = 2^x$ in order, then state the horizontal asymptote and the point corresponding to $(0, 1)$ on the parent graph.

Exercise 13 (sketch and explain). Sketch $y = e^{x-2}$ and $y = e^x - 2$ on separate axes. Explain in one sentence why these two graphs are different, even though both come from shifting $y = e^x$ by 2.

Exercise 14 (harder). A bacteria colony is modeled by $N(t) = 500 \cdot 2^{t/3}$, where t is measured in hours. Show directly that $N(t+3) = 2N(t)$, then state the horizontal asymptote and explain what it means in the context of the model. (A later logarithms lecture will show how to rewrite this model using e .)

5. Growth and Decay Models

Applications usually state a *rate per period* rather than a base. The **percentage-rate form** of an exponential model is

$$P(t) = P_0(1 + r)^t,$$

where $P_0 = P(0)$ is the initial amount and r is the rate per period, written as a decimal (positive for growth, negative for decay).

- If $r > 0$, the base $b = 1 + r > 1$: **exponential growth**, increasing by $r \times 100\%$ each period.
- If $-1 < r < 0$, the base $b = 1 + r$ satisfies $0 < b < 1$: **exponential decay**, decreasing by $|r| \times 100\%$ each period.

Example (growth). A town of 12,000 grows 3% per year: $P(t) = 12000(1.03)^t$, base $b = 1.03 = 1 + 0.03$. After 5 years, $P(5) = 12000(1.03)^5 \approx 12,000(1.1593) \approx 13,910$.

Example (decay). A car worth \$24,000 depreciates 15% per year. Losing 15% leaves 85%, so $b = 1 - 0.15 = 0.85$ and $V(t) = 24000(0.85)^t$. After 4 years, $V(4) = 24000(0.85)^4 \approx 24,000(0.5220) \approx \$12,533$.

Half-life and doubling time. Decay is often described instead by its **half-life** h , the time for the quantity to fall to half its value:

$$A(t) = A_0 \left(\frac{1}{2}\right)^{t/h}.$$

Growth is likewise described by its **doubling time** d :

$$P(t) = P_0 \cdot 2^{t/d}.$$

Both are percentage-rate models in disguise — setting $t = h$ gives $A(h) = A_0 \cdot \frac{1}{2}$, and setting $t = d$ gives $P(d) = 2P_0$, matching the name. Whenever t is a whole multiple of h or d , these evaluate without a calculator.

Example. Iodine-131 has a half-life of 8 days. Starting from $A_0 = 40$ mg, after 24 days (= 3 half-lives),

$$A(24) = 40 \left(\frac{1}{2}\right)^{24/8} = 40 \left(\frac{1}{2}\right)^3 = 40 \cdot \frac{1}{8} = 5 \text{ mg}.$$

Exercise 15 (routine). A population of 5,000 bacteria grows 8% per hour. Write $P(t)$ in percentage-rate form and find the population after 3 hours.

Exercise 16 (routine). A machine worth \$8,000 depreciates 20% per year. Write $V(t)$ in percentage-rate form, state whether this is growth or decay and why, and find its value after 3 years.

Exercise 17 (half-life). A radioactive sample has a half-life of 6 hours and starts at $A_0 = 80$ g. Find the amount remaining after 18 hours, without a calculator.

Exercise 18 (doubling time). A culture of yeast doubles every 2 hours, starting from 200 cells. Write $P(t)$ in doubling-time form, and find the population after 8 hours.

Exercise 19 (model from data). A culture has 300 cells at $t = 0$ and 1200 cells at $t = 4$ hours. Assume $P(t) = ab^t$.

- Find a and b .
- Write the model $P(t)$.
- Use it to predict the population after 6 hours.

Exercise 20 (challenge). Show that the percentage-rate form $P(t) = P_0(1 + r)^t$ and the doubling-time form $P(t) = P_0 \cdot 2^{t/d}$ describe the same function exactly when $1 + r = 2^{1/d}$ — that is, a growth rate r per period corresponds to doubling time d .